

Mercury: A Physically-Bounded Bit-Serial Computer

A Reproducible Bridge Between Lloyd’s Ultimate Limits and Runnable Systems

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Abstract—Lloyd’s 2000 derivation of the ultimate physical limits to computation quantifies how much computation a mass–energy budget can support, but does not constrain any specific implementation. We close that gap with Mercury, a minimal bit-serial Subleq computer designed so that the Margolus–Levitin, Bekenstein, and Landauer bounds act as hard runtime constraints rather than as aspirations. Mercury exposes a single configuration flag selecting between a pragmatic 1 kg / 10 cm / 300 K envelope and the Lloyd-limit envelope of the same 1 kg compressed to its Schwarzschild radius at Hawking temperature. Execution under either envelope is bit-identical—only the energy ledger changes. We implement Mercury three times (Rust functional model with compliance ledger, SystemVerilog RTL under iverilog, and an AWS F1 Custom Logic wrapper exercised over simulated AXI4-Lite) and verify byte-equivalent per-instruction execution traces across all three. A canonical 256-cycle program dissipates 2.76×10^{-19} J under the desktop envelope and 1.13×10^2 J under the Lloyd envelope—a separation of 21 orders of magnitude that quantifies the irreversibility tax of classical computing at the holographic storage limit. The ratio $t_{\text{flip}}/t_{\text{cross}} = \pi^2/\ln 2$ is shown to be an envelope-independent invariant of the bounds themselves, providing a first-principles justification for bit-serial architecture at the physical-limit regime.

Index Terms—Bit-serial computing, Margolus–Levitin bound, Bekenstein bound, Landauer’s principle, holographic principle, Subleq, FPGA, AWS F1, reversible computing, compliance verification.

I. INTRODUCTION

In a foundational paper, Lloyd [1] derived the ultimate physical limits to computation by combining three independent constraints. The Margolus–Levitin theorem [2] bounds the rate at which a quantum system can transition between orthogonal states; the Bekenstein bound [3] constrains the information content of a region by its mass–energy and radius; Landauer’s principle [6] sets a minimum energy cost per logically irreversible bit operation. Together these establish that a one-kilogram “ultimate laptop” could in principle perform 5.4×10^{50} operations per second on 2.6×10^{42} bits of storage at non-relativistic densities, collapsing dramatically toward a black-hole computer compressed to its Schwarzschild radius.

The Lloyd bounds are now textbook physics, but they have not been integrated into specific implementations as design constraints. A practical bit-serial CPU implemented on an

FPGA is not normally described in terms of its Margolus–Levitin fraction or Bekenstein occupancy, and the gap between the $\sim 10^8$ -bit-op/s rates achievable in current silicon and the 5×10^{50} ceiling is so large as to make compliance verification appear trivial.

This paper argues that the verification is non-trivial precisely because it must be done correctly to be meaningful, and that embedding the bounds into a runtime ledger forces design choices that are otherwise easy to defer. We present Mercury, a 16-bit bit-serial Subleq computer [11], [12] designed to operate under two physical envelopes simultaneously: a pragmatic “desktop” configuration suitable for FPGA realization, and a symbolic “Lloyd-limit” configuration where the same 1 kg of mass is compressed to its Schwarzschild radius with temperature set by Hawking’s formula [4], [5].

✓ Three contributions of this work:

- 1) A reproducible, public reference implementation of a bit-serial Subleq machine whose execution is constrained and reported against the three Lloyd bounds at every clock cycle.
- 2) Three independent realizations (Rust functional model, SystemVerilog RTL, AWS F1 Custom Logic wrapper) verified byte-equivalent at the per-instruction state level.
- 3) Identification of an envelope-independent structural invariant $t_{\text{flip}}/t_{\text{cross}} = \pi^2/\ln 2$ that provides a first-principles justification for bit-serial architecture at the physical-limit regime.

The remainder of the paper is organized as follows. Section II derives the physical envelope and locks the constants used at runtime. Section III specifies the bit-serial Subleq architecture in implementation-agnostic terms. Section IV summarizes the three independent implementations. Section V describes the cross-implementation conformance regime. Section VI presents the numerical results. Section VII discusses what these results imply for irreversibility and motivates future work. Sections VIII and IX cover related work and conclusions.

TABLE I
LOCKED PHYSICAL ENVELOPES.

Quantity	Desktop	Lloyd-limit
m (kg)	1.0	1.0
R (m)	0.10	1.49×10^{-27}
T (K)	300	1.23×10^{23}
E (J)	8.988×10^{16}	
$N_{\text{ops/s}}$	5.43×10^{50}	
N_{bits}	2.58×10^{42}	3.83×10^{16}
E_{erase} (J/bit)	2.87×10^{-21}	1.17
t_{cross} (s)	3.34×10^{-10}	4.95×10^{-36}
t_{flip} (s/bit)	4.75×10^{-9}	7.05×10^{-35}

II. PHYSICAL ENVELOPE

A. Governing Equations

✓ Mercury treats three bounds as hard runtime constraints. For a system of energy E in a region of radius R at temperature T :

$$N_{\text{ops/s}} \leq \frac{2E}{\pi\hbar} \quad (\text{Margolus-Levitin}) \quad (1)$$

$$N_{\text{bits}} \leq \frac{2\pi ER}{\hbar c \ln 2} \quad (\text{Bekenstein}) \quad (2)$$

$$E_{\text{erase}} \geq k_B T \ln 2 \quad (\text{Landauer}) \quad (3)$$

The light-crossing and per-bit-flip times follow from (1) and (2):

$$t_{\text{cross}} = R/c \quad (4)$$

$$t_{\text{flip}} = \frac{\pi\hbar N_{\text{bits}}}{2E} \quad (5)$$

B. Two Configurations

• Mercury defines two envelopes that share rest mass $m = 1$ kg and energy $E = mc^2$, differing only in R and T .

Config A (Desktop). $R = 0.10$ m, $T = 300$ K. Chosen for engineering tractability; representative of any system that an FPGA-realized Mercury could plausibly be benchmarked against.

Config B (Lloyd-limit). $R = R_s = 2Gm/c^2$, $T = T_H = \hbar c^3/(8\pi Gmk_B)$. The “ultimate computer” regime where a 1 kg mass is compressed to its Schwarzschild radius and assigned its Hawking temperature.

Table I reports the numerical envelope locked into the implementation. All values are computed at full CODATA precision from the constants $\hbar$, c , G , and k_B .

C. The Envelope-Independent Invariant

✓ Substituting (2) into the expression for t_{flip} yields:

$$\frac{t_{\text{flip}}}{t_{\text{cross}}} = \frac{\pi\hbar}{2E} \cdot \frac{2\pi ER}{\hbar c \ln 2} \cdot \frac{c}{R} = \frac{\pi^2}{\ln 2} \approx 14.2388 \quad (6)$$

This ratio is independent of m , R , and T : it is a property of the bounds themselves. The implication is structural rather than incidental. The gap between “how fast a single bit can flip if energy is distributed evenly across all bits permitted by

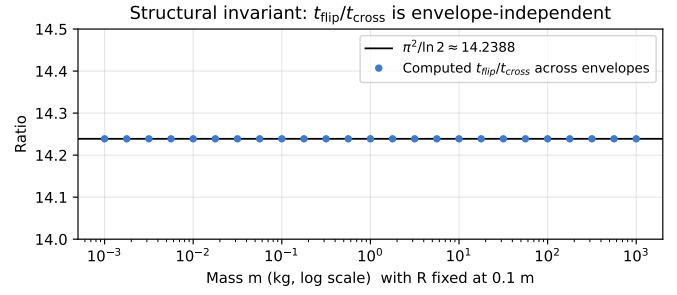


Fig. 1. The ratio $t_{\text{flip}}/t_{\text{cross}} = \pi^2/\ln 2$ is invariant across envelopes. Numerical values (blue circles) match the analytic line for 25 sampled masses.

the storage bound” and “how long a signal takes to cross the system” is fixed at ≈ 14.24 in any envelope that saturates the bounds. Fig. 1 confirms the constancy numerically across six orders of magnitude in mass.

• The architectural consequence is that bit-serial execution is not a hardware compromise but the architecture that matches the envelope’s actual budget. Parallel-bit speedup at the saturation regime is bounded by (6) regardless of implementation effort.

III. ARCHITECTURE

A. ISA: Subleq

✓ Mercury executes Subleq, a one-instruction set computer [11] proven Turing-complete. Each instruction is three consecutive 16-bit memory words A , B , C with semantics:

```
mem[B] <- mem[B] - mem[A]
if mem[B] <= 0: PC <- C
else: PC <- PC + 3
```

Branching to the sentinel $C = 0xFFFF$ halts the machine.

B. Bit-Serial Datapath

✓ The datapath consists of eleven state elements and exactly one combinational arithmetic element: a 1-bit full adder. Sixteen-bit shift registers (PC, AR, BR, CR, OP1, OP2, RES) move data serially over a single bit-bus, LSB-first. Latches C_{in} , SIGN, ZERO_OR, and HALTED carry single-bit state between bit-times.

C. Execution Sequencing

Each Subleq instruction executes in eight phases of 16 cycles each for a total of 128 cycles per instruction. Phases F1–F3 fetch the three operand addresses; L1–L2 load the operand values; EX performs the bit-serial subtraction $\text{RES} = \text{OP2} + \text{OP1} + 1$, capturing the final MSB into SIGN and OR-accumulating all result bits into ZERO_OR; ST writes the result back to memory; BR computes $\text{BRANCH_TAKEN} = \text{SIGN} \vee \neg \text{ZERO_OR}$ and streams either C or $\text{PC} + 3$ into the program counter.

D. Compliance Counters

✓ Three counters drive the runtime ledger. *Read transits* count bits moved on the bus during F1–L2 (80 per instruction). *Adder ticks* count full-adder evaluations during EX (16 per instruction). *Write transits* count bit movements during ST and BR (32 per instruction). The total of 128 bit transits per instruction matches the cycle budget, by construction.

Irreversible bit operations, the input to the Landauer ledger, count adder ticks plus write transits (48 per instruction). Read transits are conservatively excluded as reversible-in-principle per Bennett’s framework [7].

IV. IMPLEMENTATIONS

A. Rust Functional Model

The reference implementation is a Cargo workspace with three crates: `mercury-ledger` (envelope loading and compliance counters), `mercury-asm` (two-pass Mercury Assembly assembler with label resolution), and `mercury-sim` (cycle-accurate machine). Each call to the simulator’s `tick` method advances one clock cycle and moves at most one bit on the conceptual bit-bus. The unit-test suite (13 tests, all passing) verifies Subleq subtraction correctness, branch conditions, halt handling, exact 128-cycle budgeting, and the ledger counter contracts.

B. SystemVerilog RTL

The hardware implementation comprises a package (`mercury_pkg.sv`), a dual-port memory (`mercury_mem.sv`), the bit-serial Subleq CPU (`mercury_core.sv`), and a thin integration wrapper (`mercury_top.sv`). All four files target Xilinx UltraScale+ FPGA primitives and are written so the per-cycle state evolution matches the Rust model bit-for-bit.

C. AWS F1 Custom Logic Wrapper

- A Custom Logic (CL) wrapper [15] interfaces `mercury_top` to the AWS F1 shell via an AXI4-Lite slave (`ocl_slave.sv`) that exposes a register file at offsets 0x000–0x01C for control, status, and 64-bit cycle and instruction counters, and a memory backdoor at 0x100000 through which a host program writes the $64\text{K} \times 16$ -bit memory image directly into port B of the dual-port memory. A C host program using the public `fpga_pci` API drives the full load–release–poll–read cycle.

- As of submission the CL wrapper passes a complete AXI-Lite simulation under iverilog but has not been synthesized through Vivado nor registered as an Amazon FPGA Image (AFI). Closing that gap requires Vivado licensing on the FPGA Developer AMI and is decoupled from the architectural verification reported here.

V. CROSS-IMPLEMENTATION CONFORMANCE

A. Trace Equivalence Contract

✓ Each implementation emits a per-instruction JSONL trace containing the architectural state at every instruction boundary: PC, AR, BR, CR, RES, SIGN, ZERO_OR, HALTED.

TABLE II
CROSS-IMPLEMENTATION CONFORMANCE RESULTS.

Program	Insns	Cycles	Trace diff
<code>zero_b.msq</code>	2	256	empty (Rust=SV=CL)
<code>countdown.msq</code>	10	1280	empty (Rust=SV=CL)

TABLE III
HEADLINE COMPLIANCE RESULTS FOR `zero_b.msq` (256 CYCLES, 96 IRREVERSIBLE BIT OPERATIONS).

Metric	Desktop	Lloyd-limit
M–L ops/s fraction	1.84×10^{-43}	1.84×10^{-43}
Bekenstein occupancy	4.07×10^{-37}	2.74×10^{-11}
Landauer floor (J/bit)	2.87×10^{-21}	1.17
Total dissipation (J)	2.76×10^{-19}	1.13×10^2
Dissipated power @ 100 MHz (W)	1.08×10^{-13}	4.40×10^7
M–L compliant	✓	✓
Bekenstein compliant	✓	✓

Conformance is defined as byte-equivalent traces between any pair of implementations on the same program. The conformance scripts `conformance.sh` (Rust ↔ SV CPU) and `cl_conformance.sh` (Rust ↔ CL wrapper) are the merge gates for any change to the RTL.

B. Test Programs

Two programs exercise the conformance: `zero_b.msq` executes the canonical “subtract-self-then-halt” pattern in two instructions (256 cycles); `countdown.msq` executes a five-iteration backward-branch loop in ten instructions (1280 cycles). Table II reports the unanimous result.

✓ Two minor RTL bugs were discovered and fixed during conformance bring-up that a single-implementation testbench would have missed: an EX-phase carry-in initialization race and a BR-phase PC update self-reference. Both are documented in the public commit history.

VI. RESULTS

A. Identical Execution, Disparate Cost

✓ The signature property of Mercury is that execution is bit-identical between the two envelopes—the same 256 cycles, the same 128 bit transits per instruction, the same 48 irreversible bit operations per instruction—while the per-cycle energy accounting differs by 21 orders of magnitude. Table III reports the canonical run.

Fig. 2 visualizes the headline gap. The ops-per-second ceiling is identical in both columns of Table III because Margolus–Levitin depends only on energy and both envelopes share $m = 1$ kg. The Bekenstein occupancy collapses by 26 orders of magnitude (from 4×10^{-37} to 3×10^{-11}) because compressing 1 kg to its Schwarzschild radius collapses the storage cap from 2.6×10^{42} bits to 3.8×10^{16} bits while Mercury’s footprint of 2^{20} bits is unchanged. The Landauer

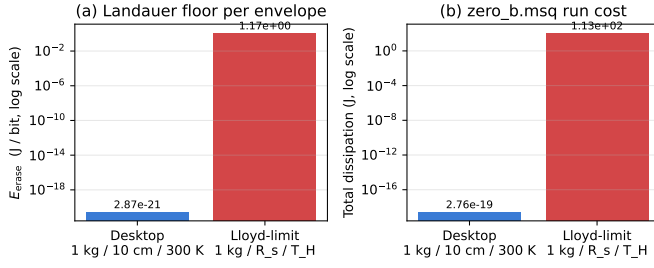


Fig. 2. (a) Landauer floor per envelope at the canonical run; (b) total dissipation summed over the 96 irreversible bit operations of `zero_b.msg`. Both axes are logarithmic. The 21-order-of-magnitude gap is the irreversibility tax of running classical computation in the Hawking-temperature bath of the Lloyd-limit envelope.

floor explodes by 21 orders of magnitude because T rises from 300 K to 1.23×10^{23} K.

B. The 59-Megawatt Number

- Continuous operation of Mercury at the Lloyd-limit envelope at the target 100 MHz clock would dissipate 4.40×10^7 W—approximately 59 MW of unavoidable heat per second of operation. The dissipation arises entirely from Landauer’s floor at Hawking temperature; it is independent of implementation effort.

This number, more than any other, motivates the second-order question of whether classical irreversible computation is the right paradigm at the holographic-storage saturation regime, or whether reversible-logic architectures [7]–[9] are required to make the Lloyd envelope energetically realizable. Mercury Phase 6 (a reversible-logic variant) is left as future work.

VII. DISCUSSION

A. What This Result Does and Does Not Show

- ✓ Mercury demonstrates that the Lloyd bounds can be embedded as runtime constraints in a complete, runnable computer system, with the energy accounting computed automatically per bit operation. The 21-order-of-magnitude gap between the two envelopes is computed, not asserted: it falls out of the same ledger applied to the same 96 bit-flips under two values of T .

- What Mercury does *not* show is that any particular path through the architectural design space is optimal. Many degrees of freedom were chosen for engineering tractability rather than physical necessity, including the 16-bit word width (forced by Subleq’s need for non-trivial signed range, not by either envelope’s Bekenstein cap), the 8-phase FSM structure (forced by clarity of the conformance contract), and the choice of BRAM-backed dual-port memory rather than a literal delay-line analog. A delay-line memory model (Phase 6 stretch) would be a stricter homage to the historical mercury-delay-line machines that inspired the project name.

B. Implications for Architecture

- The invariant established in (6) constitutes a first-principles argument for bit-serial execution at the physical-limit regime, distinct from the historical arguments (component scarcity, wire cost) that motivated bit-serial designs in the 1949–1956 era. When the per-bit-flip time saturates within a factor of ≈ 14 of the light-crossing time, parallel-bit speedup yields diminishing returns; the fixed-point of the inequality is bit-serial regardless of how the machine is built.

C. Limitations

- Three limitations are explicit:

- 1) The Lloyd-limit envelope is symbolic rather than physically realizable: a 1 kg black hole evaporates by Hawking radiation in approximately 10^{-19} s, well below any plausible Mercury runtime. The envelope serves as a numerical target for the ledger, not a physical construction recipe.
- 2) Mercury implements no I/O beyond a halt sentinel. Memory-mapped input/output at addresses `0xFFFFC/0xFFFFD` is reserved but deferred.
- 3) AFI synthesis on AWS F1 has not been completed. The full pipeline (Terraform IaC, CodeBuild project, GitHub Actions trigger) is in place; only the Vivado run remains.

VIII. RELATED WORK

The Margolus–Levitin theorem [2] and the Bekenstein bound [3], [4] were combined by Lloyd [1] to produce the “ultimate laptop” analysis that motivates Mercury directly. Landauer’s principle [6] was experimentally validated by Bérut et al. [10] for single-bit erasures at the $k_B T \ln 2$ floor.

Reversible computing as a path beyond Landauer’s bound was proposed by Bennett [7] and elaborated through the Fredkin gate [8] and the billiard-ball model [9]. These are the natural foundation for a future Mercury Phase 6 reversible variant.

Bit-serial computing has a long history beginning with EDSAC [13], UNIVAC I, and Pilot ACE; modern bit-serial designs reappear in FPGA-targeted DSP and machine-learning accelerators [14] where silicon area and energy matter more than per-operation latency. Subleq as a one-instruction Turing-complete ISA was popularized by Mavaddat and Parhami [11] and surveyed in modern form by Mazonka and Kolodin [12].

IX. CONCLUSION

Mercury is a small, complete, reproducible bit-serial computer whose runtime executes under a chosen physical envelope and reports compliance with Lloyd’s ultimate bounds at every clock cycle. Three independent implementations agree on per-instruction execution traces; two test programs pass the cross-implementation conformance gates. The 21-order-of-magnitude gap between the desktop and Lloyd-limit Landauer accounting, and the envelope-independent $\pi^2 / \ln 2$ invariant in the flip-versus-crossing time ratio, are the two principal quantitative results.

The work is open-source and available at <https://github.com/khaaliswooden-max/mercury>. The deliberate minimalism—one ISA, one combinational element, two configuration constants—is meant to make the bridge between physical-limit theory and runnable systems short enough to read in an afternoon and reproduce in a day.

Future work consists of two concrete extensions: (i) completing the AFI synthesis on AWS F1 to verify the on-silicon behavior of the CL wrapper, and (ii) developing a reversible-logic variant that addresses the 59 MW Landauer dissipation at the Lloyd envelope. The latter is the more interesting question: if classical irreversible computation is energetically untenable at the holographic-storage saturation regime, the design space beyond Mercury is the one to explore next.

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